PRODUCT REQUIREMENT DOCUMENT (PRD)

Ta-Daa AI: snap, shop & cook

DOCUMENT VERSION v1.1 (MVP Specification)	**TARGET PLATFORMS** iOS & Android (React Native / Flutter)
DESIGN TOOL TARGET Google Stitch / UI Generation Frameworks	**BACKEND INFRASTRUCTURE** Firebase (Auth, Firestore, Functions, Storage)

1. Executive Summary & Vision

Ta-Daa AI: snap, shop & cook is an intelligent mobile culinary companion designed to solve daily food indecision and simplify cooking. Users can capture a photo of a dish or describe what they crave in text. The app identifies the recipe, breaks down required ingredients, locates missing items at nearby stores, and guides the user through cooking with interactive step-by-step instructions.

2. Tiered Feature Access Model

To support monetization while maximizing acquisition, features are split into Free and Premium tiers:

Feature Module	Tier	Description & Logic
Text Dish Search	FREE	Describe desired food via text input (e.g., "creamy garlic pasta").
Photo Recognition	PREMIUM	Snap or upload a photo of food to auto-extract recipe & ingredients.
Local Grocery Finder	FREE	Locate nearby supermarkets for missing ingredients via Google Places.
Guided Cooking Steps	FREE	Step-by-step text view with visual timers and checkable steps.
Voice Assistant Mode	PREMIUM	Hands-free voice guidance and streaming AI Q&A while cooking.

3. Screen-by-Screen UI & Layout Specifications for Google Stitch

Below are detailed layout requirements for all key screens to feed directly into Google Stitch or UI design prompts.

Screen 1: Home / Dish Discovery Hub

Central entry point for searching meals or capturing dish photos.

Key UI Components & Elements:

- **Top Bar:** App Logo (Ta-Daa AI), Premium Status Pill Badge, User Avatar.
- **Primary Hero Section:** Search bar ("What do you want to eat today?").
- **Camera Capture Button (Premium):** Prominent camera floating icon or card labeled "Snap Your Food" with a subtle lock indicator if user is on Free tier.
- **Quick Inspiration Chips:** Horizontal scroll chips for quick ideas ("Under 20 Mins", "High Protein", "Italian", "Vegetarian").
- **Recent Meals Grid:** 2-column card layout showing saved or recently cooked dishes.

Screen 2: Paywall & Subscription Upgrade Modal

Triggered when a Free user attempts to open the Camera scanner or Voice Assistant.

Key UI Components & Elements:

- **Hero Image / Illustration:** Vibrant graphic showing camera AI scanning food and voice cooking.
- **Headline:** "Unlock Food Snap & Hands-Free Voice Cooking".
- **Feature Comparison Checklist:**
  - ✓ Snap any dish to extract recipe instantly
  - ✓ Hands-free streaming voice assistant while cooking
  - ✓ Unlimited recipe conversions & pantry sync
- **Pricing Plan Switcher:** Monthly ($4.99/mo) vs Annual ($39.99/yr - Save 33%).
- **CTA Button:** "Start 7-Day Free Trial".
- **Footer Links:** Restore Purchases | Terms of Service | Privacy Policy.

Screen 3: Ingredient Checklist & Store Map

Displays extracted ingredients and identifies nearby stores for missing items.

Key UI Components & Elements:

- **Header:** Dish Name, Servings Stepper, Total Time, Estimated Cost.
- **Pantry Check Section:** List of ingredients with checkboxes ("I have this"). Items checked move to a "Pantry" collapsed section.
- **Missing Items Callout:** Highlighted list of items needing purchase.
- **Map View Container:** Embedded map widget (Google Maps/Mapbox styling) showing user location pinned with top 3 nearest supermarkets.
- **Store Cards:** Horizontal carousel at bottom showing Store Name, Distance (e.g., 1.2 km), Rating, and "Get Directions" button.
- **Bottom Sticky CTA:** "Start Cooking Now" button.

Screen 4: Interactive Step-by-Step Cooking Mode

Guided interface for preparing the meal in the kitchen.

Key UI Components & Elements:

- **Top Progress Bar:** Step indicator (e.g., Step 3 of 7 - 42% complete).
- **Current Step Card:** Large bold typography explaining current action clearly.
- **Embedded Timer Widget:** Countdown timer with Start/Pause/Reset controls if step contains a duration.
- **Voice Controls Bar (Premium Only):** Live audio wave animation with mic toggle button indicating "Listening for 'Next Step' or questions...".
- **Navigation Controls:** Large bottom navigation buttons ("Previous", "Next Step").

4. Technical & API Integration Requirements

- **Authentication & Profile:** Firebase Auth (Google Sign-In, Email/Password, Apple ID).

- **Subscription Management:** RevenueCat SDK integration for Apple IAP and Google Play Billing. Firestore synced via RevenueCat Firebase Extension to maintain `is_premium` user flag.
- **AI Vision & Multimodal Processing:** Google Gemini 1.5 Flash API or OpenAI GPT-4o Vision API. Structured JSON output for recipe breakdown.
- **Location & Mapping API:** Google Places API (Nearby Search & Distance Matrix) filtered by `supermarket` or `grocery_or_supermarket`.
- **Voice Interface (Premium):** WebRTC / WebSocket streaming connection using OpenAI Realtime API or ElevenLabs Conversational AI SDK.

5. Master Prompt Template for Google Stitch

GOOGLE STITCH / AI UI DESIGN PROMPT

Create a modern, clean, and vibrant mobile app UI design system for a smart cooking application named "Ta-Daa AI: snap, shop & cook". Use a clean palette of warm emerald green (#10B981), warm amber (#F59E0B), and crisp neutral light backgrounds (#FAF9F6).

Please generate the following 4 core screens with high visual fidelity and mobile touch ergonomics:
1. **Home Screen:** Search bar, "Snap Your Food" camera banner with a premium lock badge, fast feature chips, and 2-column food card recommendations.
2. **Paywall Screen:** High-converting subscription modal presenting Monthly/Annual options, feature checkmarks (Photo Snap & AI Voice Mode), and a primary "Start Free Trial" CTA button.
3. **Ingredient & Store Finder:** Ingredient checklist with toggle boxes, "Pantry" accordion, and an integrated bottom map showing nearby supermarkets with distance pins.
4. **Cooking Assistant Screen:** Step-by-step full screen guide with step progress bar, clear step description, countdown timer widget, and a bottom streaming voice wave indicator for hands-free mode.